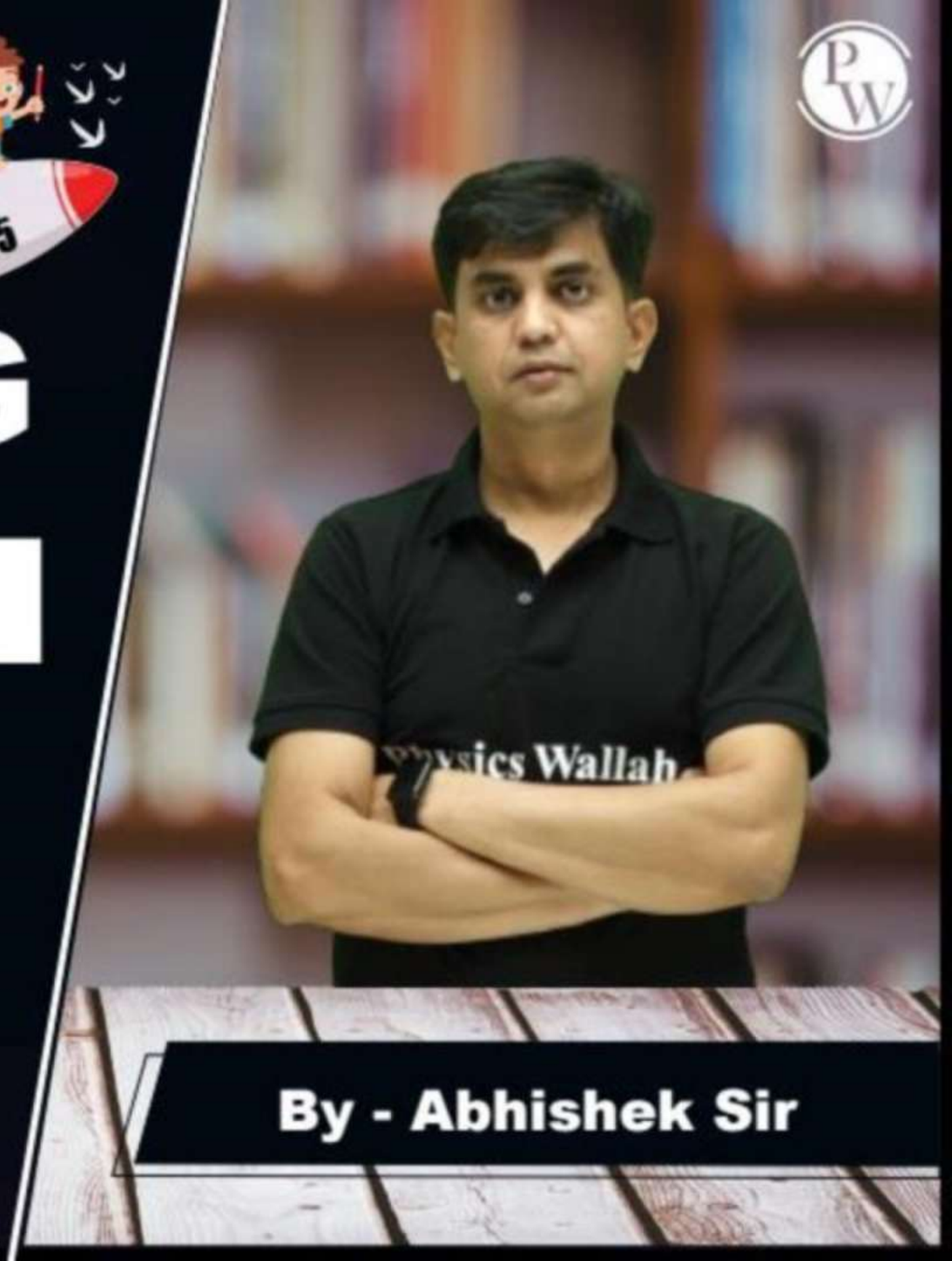
CS & IT ENGINEERING

Computer Network

Transport Layer

Lecture No. - 09



By - Abhishek Sir

Recap of Previous Lecture

Topic TCP Congestion Control

Topics to be Covered

Topic TCP Congestion Control

Topic TCP Timers

ABOUT ME

Hello, I'm Abhishek

- GATE CS AIR - 96
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#Q. On a TCP connection, current congestion window size is Congestion Window = 4 KB. The window size advertised by the receiver is Advertise Window = 6 KB. The last byte sent by the sender is LastByteSent = 10240 and the last byte acknowledged by the receiver is LastByteAcked = 8192. The current window size at the sender is _____ .

[GATE-2005]

(A) 2048 bytes

(B) 4096 bytes

(C) 6144 bytes

(D) 8192 bytes

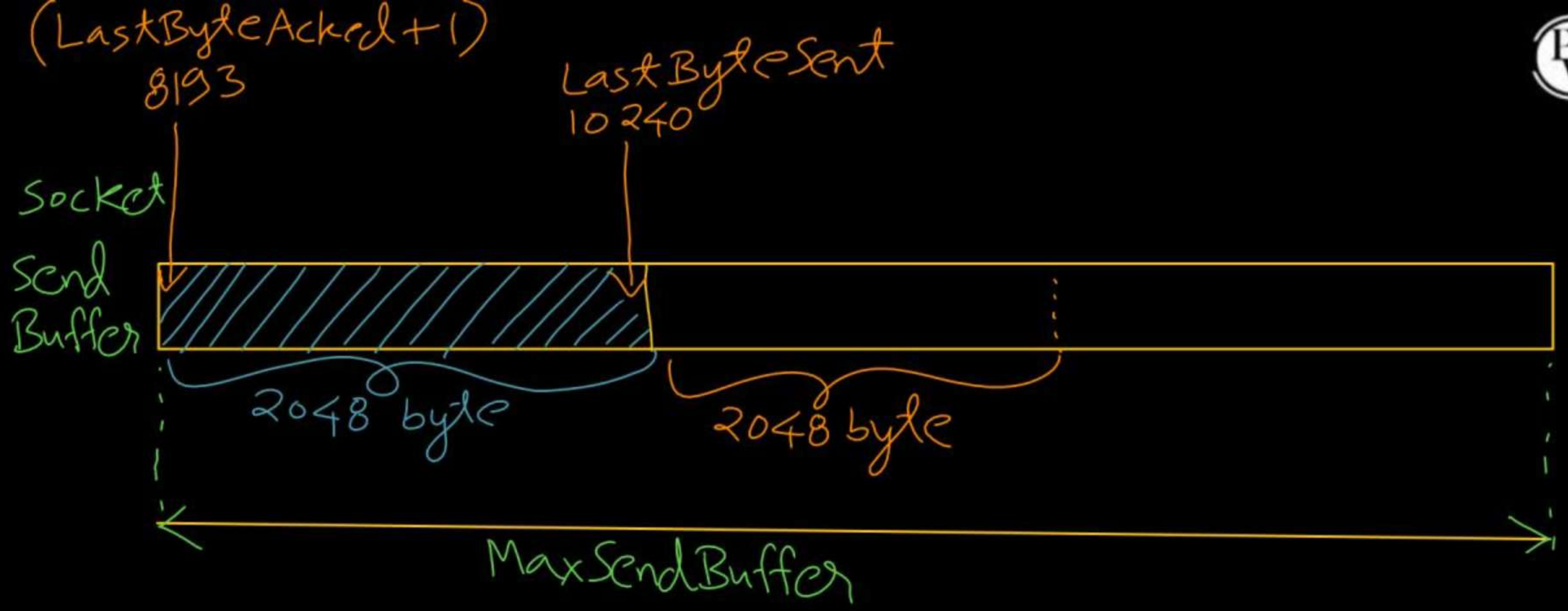
CWND = 4KB, RWND = 6KB

LastByteSent = 10240

LastByteAcked = 8192

$(\text{LastByteSent} - \text{LastByteAcked}) \leq \min(\text{CWND}, \text{RWND})$

$(10240 - 8192) \leq \min(4\text{KB}, 6\text{KB})$



$$\text{Max}^{m}\text{ Sending Limit (Window Size)} = \min(CWND, RWND)$$

#Q. Consider the following statements regarding the slow start phase of the TCP congestion control algorithm. Note that cwnd stands for the TCP congestion window and MSS denotes the Maximum Segment Size.

i. The cwnd increase by 2 MSS (1 MSS) on every successful acknowledgement. FALSE

ii. The cwnd approximately doubles on every successful acknowledgedment. FALSE

iii. The cwnd increase by 1 MSS every round trip time. FALSE

iv. The cwnd approximately doubles every round trip time. TRUE

Which one of the following is correct?

[GATE-2018, 1-Mark]

(A) Only (ii) and (iii) are true

(B) Only (i) and (iii) are true

(C) Only (iv) is true

(D) Only (i) and (iv) are true

Ans: C

#Q. Let the size of congestion window of a TCP connection be 32 KB when a timeout occurs. The round trip time of the connection is 100 msec and the maximum segment size used is 2 KB. The time taken (in msec) by the TCP connection to get back to 32 KB congestion window is _____ .

[GATE-2014, Set-1, 2-Mark]

IIT-KGP

$CWND = 32KB$

$= \frac{32KB}{2KB} = 16MSS$ *

Timeout

$SSThresh = \frac{CWND}{2}$

$= \frac{16MSS}{2} = 8MSS$

$CWND = 1MSS$

$CWND = 1MSS$ → 1

2MSS → 2

4MSS → 3

8MSS → 4

9MSS → 5

10MSS

⋮

15MSS → 11

16MSS

$RTT = 100ms$

$MSS = 2KB$

$\boxed{Ans = 11 * RTT = 1100ms}$

$\boxed{1100 - 1300}$

#Q. Consider a TCP connection between a client and a server with the following specifications; the round trip time is 6 ms, the size of the receiver advertised window is 50 KB, slow-start threshold at the client is 32 KB, and the maximum segment size is 2 KB. The connection is established at time t=0. Assume that there are no timeouts and errors during transmission. Then the size of the congestion window (in KB) at time t+60 ms after all acknowledgments are processed is _____

IIT-D

[GATE-2020, 2-Mark]

RWND = 50 KB

$= \frac{50\,KB}{2\,KB} = 25\,MSS$

SSThresh = 32 KB

$= \frac{32\,KB}{2\,KB} = 16\,MSS$

CWND = 1 MSS → 1 (t = 0)

2 MSS → 2

4 MSS → 3

8 MSS → 4

16 MSS → 5

17 MSS

21 MSS → 10

22 MSS (t = 60 ms)

RTT = 6 ms

MSS = 2 KB

Ans = 22 * 2 KB

= 44 KB

#Q. Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

[GATE-2012, 2-Mark]

IIT-D

(A) 8 MSS

(B) 14 MSS

(C) 7 MSS

(D) 12 MSS

SSThresh = 8 MSS

CWND = 2 MSS → (1)

4 MSS → (2)

8 MSS → (3)

9 MSS → (4)

10 MSS → (5)

Timeout

SSThresh = $\frac{CWND}{2}$

$\frac{10\,MSS}{2} = 5\,MSS$

CWND = 1 MSS → (6)

2 MSS → (7)

4 MSS → (8)

5 MSS → (9)

6 MSS → (10)

7 MSS

4 MSS → (8)

6 MSS → (9)

7 MSS → (10)

8 MSS

Marks to ALL

SSThresh = 5 MSS

CWND = 2 MSS → 6

4 MSS → 7

5 MSS → 8

6 MSS → 9

7 MSS → 10

8 MSS

4 MSS → 7

6 MSS → 8

7 MSS → 9

8 MSS → 10

9 MSS

#Q. Suppose that the maximum transmit window size for a TCP connection is 12000 bytes. Each packet consists of 2000 bytes. At some point of time, the connection is in slow-start phase with a current transmit window of 4000 bytes. Subsequently, the transmitter receives two acknowledgments Assume that no packets are lost and there are no time-outs. What is the maximum possible value of the current transmit window?

[GATE-2004]

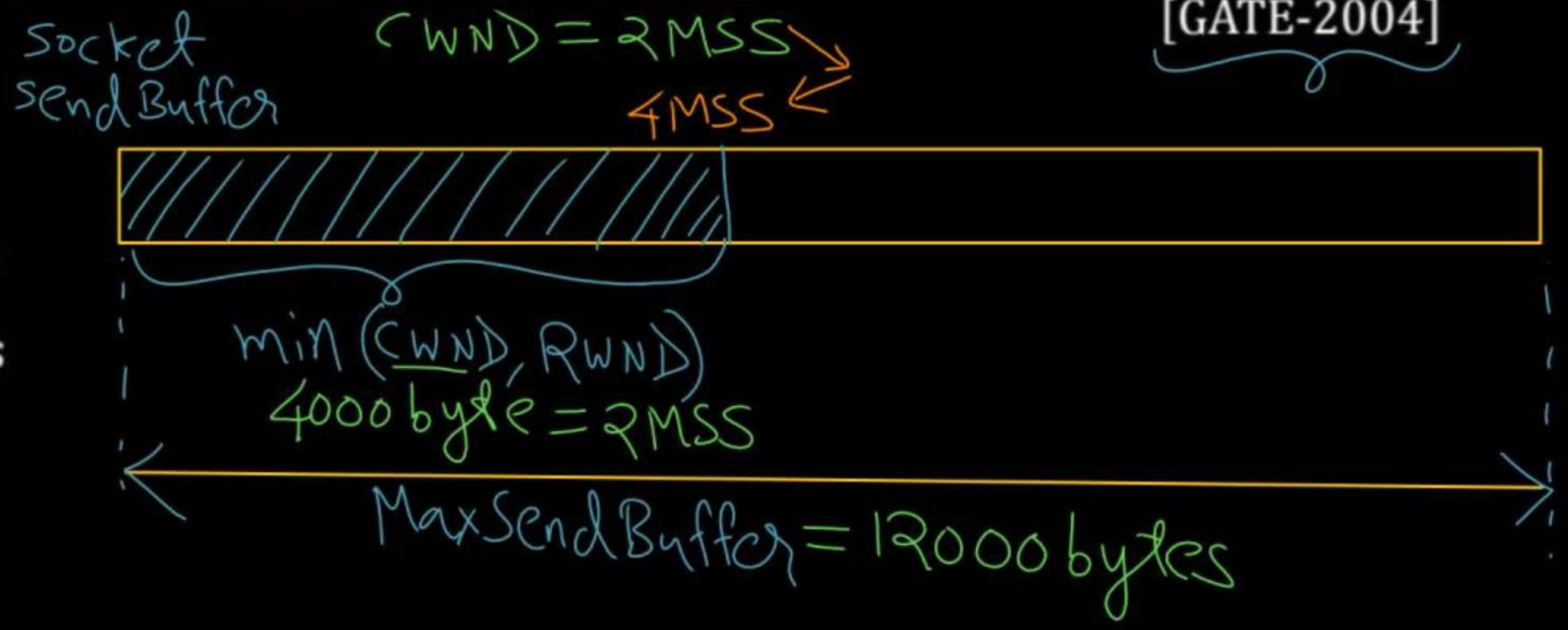
(A) 4000 bytes

(B) 8000 bytes

(C) 10000 bytes

(D) 12000 bytes

Ans: B



Topic : Retransmission Policy of TCP

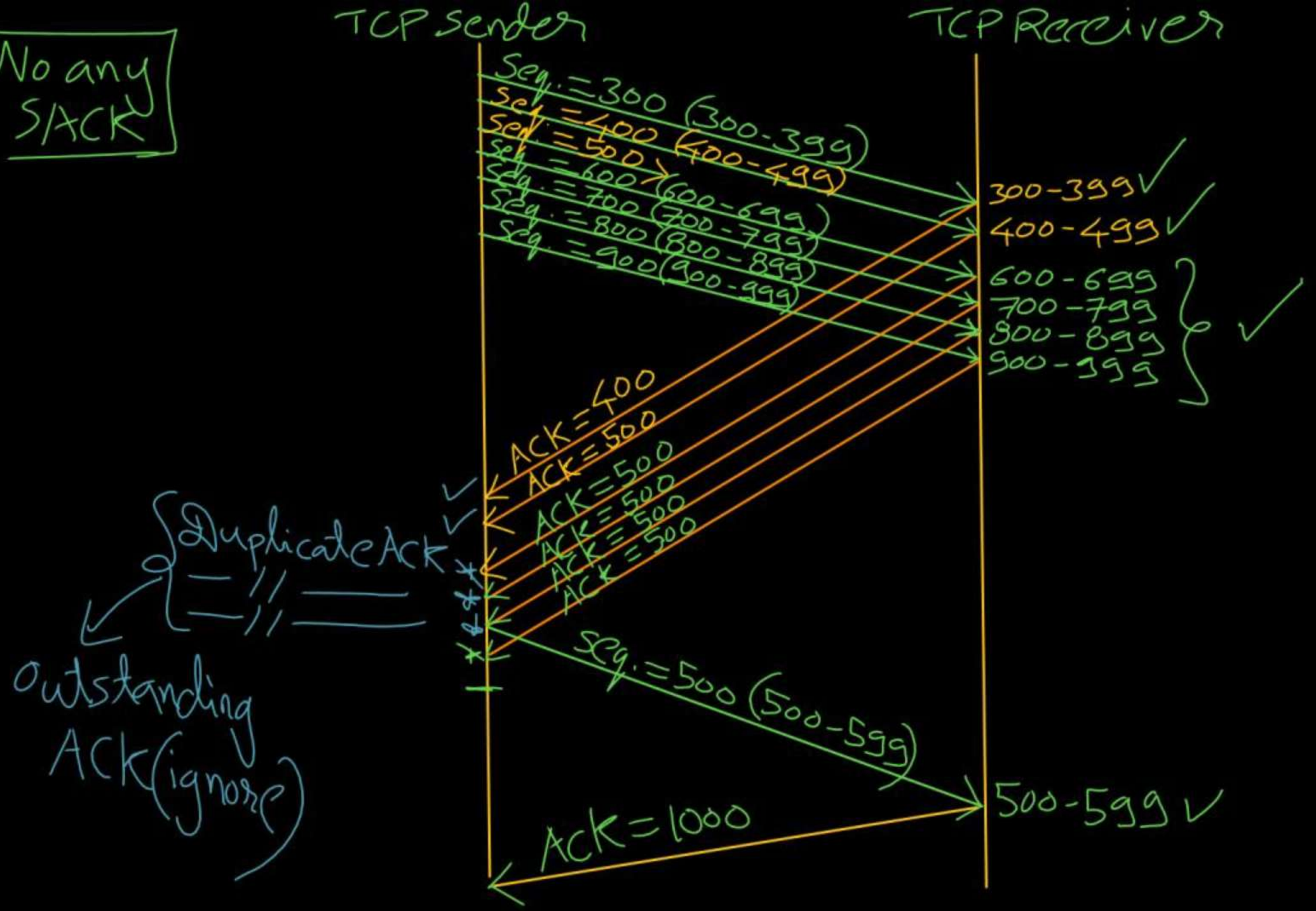
→ Retransmission on time-out
[Retransmit that segment for which time-out occur]

Timeout
SSThresh = CWND/2
CWND = 1 MSS

→ Retransmission on receiving '3 duplicate acknowledgments'
[Retransmit the unacked segment with smallest sequence number]
[Do not wait for time-out, "Fast Retransmit"]

[Early Retransmit]

No any SACK


TCP Sender
TCP Receiver
Seq. = 300 (300-399)
Seq. = 400 (400-499)
Seq. = 500 (500-599)
Seq. = 600 (600-699)
Seq. = 700 (700-799)
Seq. = 800 (800-899)
Seq. = 900 (900-999)
300-399
400-499
600-699
700-799
800-899
900-999
ACK = 400
ACK = 500
ACK = 500
ACK = 500
ACK = 500
ACK = 500
Duplicate Ack
Outstanding ACK (ignore)
Seq. = 500 (500-599)
500-599
ACK = 1000

Topic : Congestion Control

→ When Time-out occur :

Update : SSThresh = [CWND / 2] and CWND = 1 MSS

→ In case of "Fast Retransmit" :

Update : SSThresh = [CWND / 2] and CWND = 1 MSS (default)

→ "Fast Recovery" In case of "Fast Retransmit" :

Update : SSThresh = [CWND / 2] and CWND = [CWND / 2]

2 mins Summary

Topic TCP Congestion Control

Topic ~~TCP Timers~~

THANK - YOU